



Enabling Onboard Navigation for UAVs in GPS Denied Environments

MIP400A : B.Tech Project - MidTerm Evaluation

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Goal: give a UAV a way to know its own position when GPS/GNSS is unavailable, jammed, or spoofed, using only onboard sensors and physical measurements of its surroundings.

- ❑ **This semester:** get an end-to-end navigation pipeline *working* in simulation (image/data in, position out).
- ❑ **Next semester (through Apr 2027):** make the pipeline *fast and power-efficient* (FPGA / embedded compute) enough to run onboard real flight hardware.

Core approach: Scene-Aided Navigation

Replace the satellite signal with passive measurements the aircraft can already sense, such as terrain, gravity, magnetic field, and visual scene, matched against a reference map. This turns navigation into a *physical measurement + estimation* problem, much like classical inertial guidance.



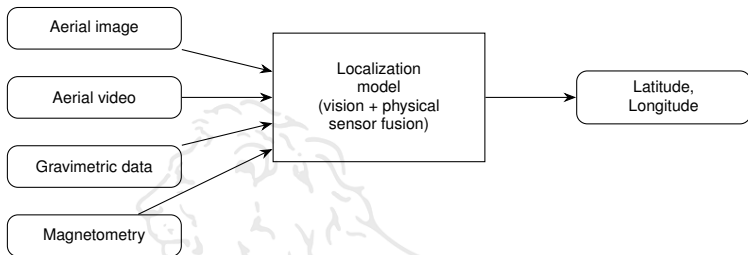
- ❑ GNSS is a **weak radio signal** broadcast from space: it is easily jammed, spoofed, or simply unavailable (urban canyons, mountains, contested airspace, indoors).
- ❑ Autonomous UAVs used in surveying, disaster response, and defense all need to keep flying *when GPS fails*, not just when it works.
- ❑ **Mechanical perspective:** a UAV is fundamentally a mechanical system in motion. Its onboard IMU already measures physical quantities such as acceleration, rotation, and gravity using mechanical/MEMS sensors. The question this project asks: can these existing physical measurements, combined with a camera, replace the satellite link entirely?
- ❑ **Precedent:** TERCOM and DSMAC, terrain and scene matching systems developed for Cold-War cruise-missile guidance, proved decades ago that passive, GPS-free navigation using physical terrain data is reliable and flight-proven.



Passive GPS-denied navigation methods, screened for feasibility on a small UAV:

Method	Physical quantity measured	Passive?	Assessment
Terrain matching (TERCOM)	Ground elevation	Yes	Proven, strong candidate ^[4,6]
Visual landmark matching	Camera imagery	Yes	Strong if paired with AI matching ^[1-3,5,7,12,13]
Transformer-based matching	Camera imagery	Yes	SOTA in this niche; motivates our redesign ^[14,15]
Gravity matching	Gravity anomalies	Yes	Promising, needs sensitive sensor ^[8,9]
Gravity-gradient nav.	$\partial g / \partial x, y, z$	Yes	Very promising, higher precision ^[8,9]
Magnetic anomaly nav.	Magnetic field	Yes	Useful in mountainous terrain ^[10,11]
Quantum inertial sensing	Acceleration / rotation	Yes	No map needed; pure mechanics
Wi-Fi / cellular / RF	Radio signal strength	No	Rejected: unavailable at altitude

Existing implementations reviewed: VGG16^[1] and CLIP-based^[2,3,7] deep feature matching of drone frames to satellite tiles (sub-3–7 m RMSE)^[2,4], classical SIFT/ORB feature matching^[5], terrain-weighted optimization against DEM maps^[4,6].



- ❑ **Current phase (no drone hardware yet):** feed pre-recorded aerial imagery into the pipeline to validate feasibility before committing to flight tests.
- ❑ **Baseline validated:** replicated the reference implementation^[1] in simulation; measured accuracy matches the numbers reported in the paper.
- ❑ **Current direction:** redesign the localization backbone as a **transformer-based** model in place of the paper's CNN features (architecture on next slide), then extend it with multimodal sensor fusion.

Proposed Model Architecture



1. **Data:** aerial images paired with high-precision ground-truth coordinates (from the replicated baseline's flight/simulation data).
2. **Stage 1 — vision backbone:** a transformer-based encoder replaces the reference paper's VGG16 features^[1], regressing image $\rightarrow$ coordinates directly.
3. **Stage 2 — multimodality:** fuse IMU, magnetometer, and gravimeter (accelerometer) vectors into the transformer alongside the visual stream; train end-to-end for high-precision localization.

Architecture diagram — to be inserted
(transformer encoder + multimodal fusion head)

Timeline of the Work



Phase 1: Now to Nov 2026 (this semester)

Reference implementation replicated and validated (done); redesign the localization backbone as a transformer model and extend it with multimodal fusion (IMU, magnetometer, gravimeter); get an end-to-end simulated pipeline (aerial image $\rightarrow$ position estimate) working.

Phase 2: Jan 2027 to Apr 2027

Optimize the model for speed and power efficiency on FPGA, extending our lab project from last semester which conclusively showed FPGAs are highly power-efficient for this class of computation; integrate onto embedded flight hardware for a physical UAV test.



- [1] H. Goforth and S. Lucey, *GPS-Denied UAV Localization using Pre-existing Satellite Imagery*, CMU Robotics Institute.
<https://publications.ri.cmu.edu/gps-denied-uav-localization-using-pre-existing-satelliteimagery>
(code: <https://github.com/hmgoforth/gps-denied-uav-localization>)
- [2] *ngps_flight*: real-time CLIP-style UAV-satellite matching, ~3 m RMSE on Jetson Orin.
https://github.com/snktshrma/ngps_flight
- [3] *CLIP-UAV-localization*: CLIP-based network matching UAV imagery, also recovers heading.
<https://github.com/codebra721/CLIP-UAV-localization>
- [4] *GNSS-Denied-UAV-Geolocalization*: terrain-weighted optimization using satellite maps and DEM data, sub-7 m.
<https://github.com/YFS90/GNSS-Denied-UAV-Geolocalization>
- [5] *VisionUAV-Navigation*: classical SIFT/ORB hybrid feature detection, no deep learning.
<https://github.com/sidharthmohannair/VisionUAV-Navigation>
- [6] GNSS-denied UAV geolocalization via terrain-weighted constraint optimization, sub-7 m error, *ScienceDirect*.
<https://www.sciencedirect.com/science/article/pii/S1569843224006332>
- [7] CLIP-based UAV-satellite image matching, also derives heading direction, *ScienceDirect*.
<https://www.sciencedirect.com/science/article/pii/S2590123025041787>





- [8] A Review of Key Technologies in Gravity Matching Navigation, *Sensors* (2026).
<https://www.mdpi.com/1424-8220/26/13/4208>
- [9] Gravity-Aided Navigation Using Viterbi Map Matching Algorithm. <https://arxiv.org/abs/2204.10492>
- [10] Absolute Positioning Using the Earth's Magnetic Anomaly Field. <https://doi.org/10.1002/navi.138>
- [11] MagNav-Based Navigation Simulation Using Magnetic Anomaly Maps for GPS-Denied Environments.
<https://doi.org/10.5302/J.ICRDS.2026.25.0344>
- [12] *PCVLabDrone2021*: real-time down-facing camera pipeline with quad-tree map retrieval, tested live.
<https://github.com/OSUPCVLab/PCVLabDrone2021>
- [13] *geolocalization*: dual/Siamese network recognizing same-scene UAV and satellite image pairs.
<https://github.com/abhinavtripathi95/geolocalization>
- [14] X. Zhu et al., *TransGeo: Transformer Is All You Need for Cross-view Image Geo-localization*, CVPR 2022.
<https://arxiv.org/abs/2204.00097>
- [15] *FSRA*: A Transformer-Based Feature Segmentation and Region Alignment Method for UAV-View Geo-Localization.
<https://arxiv.org/pdf/2201.09206>

